Muhammad Faraz Malik

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PROFESSIONAL SUMMARY

Computer Engineering student at UET Taxila with practical work in AI systems, reinforcement learning, robotics simulation, agentic AI, computer vision, networking, and automation. Experienced with Python, ROS 2, Gazebo, AirSim, LangGraph, local LLM workflows,deployment-oriented engineering.

EDUCATION

BS Computer Engineering, UET Taxila	Sep 2022 – Present | CGPA: 3.22
F.Sc. Pre-Engineering, PAEC Model College, Chashma	2020 – 2022
Matriculation, PAEC Model College, Chashma	2018 – 2020

TECHNICAL SKILLS

Programming:	Python, C++, Java, Verilog, HTML, CSS
AI/ML:	Reinforcement Learning, Q-Learning, CNNs, MFCC, STFT, agentic AI workflows, local LLM orchestration
Robotics/Simulation:	ROS 2, Gazebo, AirSim, drone simulation, autonomous navigation, MATLAB, Simulink
Tools/Systems:	LangGraph, REST APIs, Linux, Windows, OptiSystem, Arduino, AutoCAD, PDQ Deploy, Active Directory
Core Areas:	Computer vision, signal processing, networking, automation

TECHNICAL PROJECTS

AI Driving Scene Fusion and Decision System — Python, YOLOPX, Depth Anything V2, YOLO, TensorRT

- Built a multi-model video pipeline combining object detection, lane-line output, metric depth estimation, and traffic-light recognition into one fused decision layer.
- Generated annotated video plus frame-wise CSV/JSON decisions, with deployment-oriented optimization using TensorRT engines for practical inference speed.

Drone Reinforcement Learning Navigation in AirSim — Python, AirSim, RL

- Implemented simulation-based RL for autonomous drone navigation from start point to target point using state design, action selection, reward logic, and evaluation flow.

ROS 2 and Gazebo Drone Simulation Work — ROS 2, Gazebo, Python

- Worked with robotics simulation concepts including nodes, topics, simulation models, sensor feedback, and control-side integration for drone/robot experiments.

Agentic Medical AI Workflow with Local LLM — LangGraph, Local LLM, APIs, Python

- Built a six-agent LangGraph workflow around a local language model, queried two medical database APIs, and organized retrieved data into desired final outputs.

Minecraft Q-Table Reinforcement Learning Agent — Python, Q-Learning

- Developed a tabular Q-learning agent using state-action values, reward shaping, and exploration-exploitation logic to improve behavior over repeated episodes.

Speech Emotion Recognition — Python, CNN, MFCC, TESS Dataset

- Created a CNN-based emotion classification system using MFCC feature extraction to classify speech into seven emotional categories.

Remote Software Deployment Automation — PDQ Deploy, Active Directory

- Built a centralized workflow for deploying software from a server to multiple client PCs, reducing manual installation effort in a managed IT environment.

ADDITIONAL ACADEMIC WORK

MIPS Processor: Single-cycle Verilog CPU with modular datapath/control.

FSO Communication: OptiSystem simulation under weather effects.

Audio Denoising: MATLAB STFT filtering for noise reduction.

University Management System: Web platform concept for academic and administrative workflows.

LANGUAGES

English: Fluent | Urdu: Native